

6

- END OF PAPER**



[6 QER]



### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                   | NEGATIVE IONS |                    |
|---------------|-------------------|---------------|--------------------|
| Name          | Formula           | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$  | bromide       | $\text{Br}^{-}$    |
| ammonium      | $\text{NH}_4^{+}$ | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$  | chloride      | $\text{Cl}^{-}$    |
| calcium       | $\text{Ca}^{2+}$  | fluoride      | $\text{F}^{-}$     |
| copper(II)    | $\text{Cu}^{2+}$  | hydroxide     | $\text{OH}^{-}$    |
| hydrogen      | $\text{H}^{+}$    | iodide        | $\text{I}^{-}$     |
| iron(II)      | $\text{Fe}^{2+}$  | nitrate       | $\text{NO}_3^{-}$  |
| iron(III)     | $\text{Fe}^{3+}$  | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^{+}$   | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$  |               |                    |
| nickel        | $\text{Ni}^{2+}$  |               |                    |
| potassium     | $\text{K}^{+}$    |               |                    |
| silver        | $\text{Ag}^{+}$   |               |                    |
| sodium        | $\text{Na}^{+}$   |               |                    |
| zinc          | $\text{Zn}^{2+}$  |               |                    |





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

|   |   |          |   |
|---|---|----------|---|
| 1 | H | Hydrogen | 1 |
|---|---|----------|---|

|     |    |           |    |     |    |           |    |     |    |           |    |     |    |           |    |     |    |              |    |     |    |            |    |     |    |             |    |     |    |           |    |     |    |           |    |     |    |            |    |
|-----|----|-----------|----|-----|----|-----------|----|-----|----|-----------|----|-----|----|-----------|----|-----|----|--------------|----|-----|----|------------|----|-----|----|-------------|----|-----|----|-----------|----|-----|----|-----------|----|-----|----|------------|----|
| 7   | Li | Lithium   | 3  | 9   | Be | Beryllium | 4  | 11  | Na | Sodium    | 11 | 12  | Mg | Magnesium | 12 | 13  | Al | Aluminium    | 13 | 14  | Si | Silicon    | 14 | 15  | P  | Phosphorus  | 15 | 16  | S  | Sulfur    | 16 | 17  | Cl | Chlorine  | 17 | 18  | Ar | Argon      | 18 |
| 19  | K  | Potassium | 19 | 20  | Ca | Calcium   | 20 | 21  | Sc | Scandium  | 21 | 22  | Ti | Titanium  | 22 | 23  | V  | Vanadium     | 23 | 24  | Cr | Chromium   | 24 | 25  | Mn | Manganese   | 25 | 26  | Fe | Iron      | 26 | 27  | Co | Cobalt    | 27 | 28  | Ni | Nickel     | 28 |
| 37  | Rb | Rubidium  | 37 | 38  | Sr | Sr        | 38 | 39  | Y  | Yttrium   | 39 | 40  | Zr | Zirconium | 40 | 41  | Nb | Niobium      | 41 | 42  | Mo | Molybdenum | 42 | 43  | Tc | Technetium  | 43 | 44  | Ru | Ruthenium | 44 | 45  | Rh | Rhodium   | 45 | 46  | Pd | Palladium  | 46 |
| 55  | Cs | Caesium   | 55 | 56  | Ba | Barium    | 56 | 57  | La | Lanthanum | 57 | 58  | Ce | Cerium    | 58 | 59  | Pr | Praseodymium | 59 | 60  | Nd | Neodymium  | 60 | 61  | Pm | Promethium  | 61 | 62  | Sm | Samarium  | 62 | 63  | Eu | Europium  | 63 | 64  | Gd | Gadolinium | 64 |
| 87  | Fr | Francium  | 87 | 88  | Ra | Radium    | 88 | 89  | Ac | Actinium  | 89 | 90  | Th | Thorium   | 90 | 91  | Pa | Protactinium | 91 | 92  | U  | Uranium    | 92 | 93  | Np | Neptunium   | 93 | 94  | Pu | Plutonium | 94 | 95  | Am | Americium | 95 | 96  | Cm | Curium     | 96 |
| 133 | Cs | Caesium   | 55 | 137 | Ba | Barium    | 56 | 138 | La | Lanthanum | 57 | 139 | Hf | Hafnium   | 72 | 181 | Ta | Tantalum     | 73 | 182 | W  | Tungsten   | 74 | 186 | Re | Rhenium     | 75 | 187 | Os | Osmium    | 76 | 190 | Ir | Iridium   | 77 | 192 | Pt | Platinum   | 78 |
| 223 | Fr | Francium  | 87 | 226 | Ra | Radium    | 88 | 227 | Ac | Actinium  | 89 | 228 | Th | Thorium   | 90 | 232 | U  | Uranium      | 92 | 238 | Pu | Plutonium  | 94 | 244 | Cf | Californium | 98 | 254 | Am | Americium | 95 | 258 | Cm | Curium    | 96 | 262 | Bk | Berkelium  | 97 |
| 101 | Ag | Silver    | 47 | 106 | Pd | Palladium | 46 | 107 | Cu | Copper    | 29 | 108 | Ni | Nickel    | 28 | 109 | Cd | Cadmium      | 48 | 110 | In | Indium     | 49 | 111 | Sn | Stannum     | 50 | 112 | Pb | Lead      | 82 | 113 | Bi | Bismuth   | 83 | 114 | Po | Polonium   | 84 |
| 127 | I  | Iodine    | 53 | 128 | Te | Tellurium | 52 | 129 | Sb | Antimony  | 51 | 130 | Sn | Stannum   | 50 | 131 | Pb | Lead         | 82 | 132 | Bi | Bismuth    | 83 | 133 | Po | Polonium    | 84 | 134 | At | Astatine  | 85 | 135 | Rn | Radon     | 86 | 136 | Fr | Francium   | 87 |
| 201 | Hg | Mercury   | 80 | 202 | Tl | Thallium  | 81 | 203 | Pb | Lead      | 82 | 204 | Bi | Bismuth   | 83 | 205 | Po | Polonium     | 84 | 206 | At | Astatine   | 85 | 207 | Rn | Radon       | 86 | 208 | Fr | Francium  | 87 | 209 | Ac | Actinium  | 89 | 210 | Th | Thorium    | 90 |
| 285 | Fr | Francium  | 87 | 286 | Ra | Radium    | 88 | 287 | Ac | Actinium  | 89 | 288 | Th | Thorium   | 90 | 289 | Pa | Protactinium | 91 | 290 | U  | Uranium    | 92 | 291 | Np | Neptunium   | 93 | 292 | Pu | Plutonium | 94 | 293 | Am | Americium | 95 | 294 | Cm | Curium     | 96 |

Key

|        |                      |
|--------|----------------------|
| $A_r$  | relative atomic mass |
| Symbol | Name                 |
| $Z$    | atomic number        |